The Effect of Minority Faculty Representation on McNair Program Success Rates, 2010-2014

University of Minnesota

Kee Vang, McNair Scholar Dr. Samuel Myers Jr. School of Public Affairs

TRIO Ronald E. McNair Post-Baccalaureate Achievement Program

Definitions

- *Underrepresented minority faculty* are defines as: African Americans, Hispanics, American Indian, Alaskan Native, Native Hawaiian and two or more races
- *All staff* are defined as full and part time faculty, administrative, clerical, maintenance, support services,
- *Faculty* are defined as full time tenure and non-tenure professors, associate professors, assistant professors, instructors, and lecturers.

Methodology

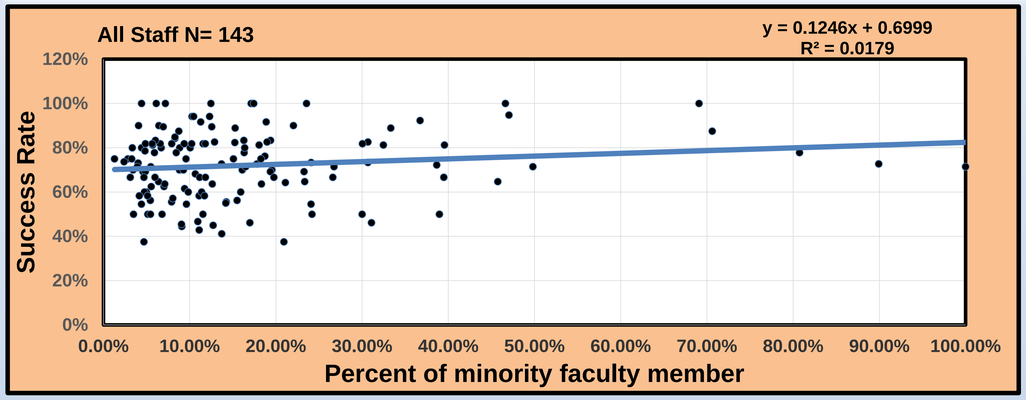
- Data from the U.S. Department of Education McNair Annual Performance Reports were obtained for 143 institutions (Research 1 = 39) We computed the McNair Scholar Success rates by finding the ratio of the McNair Scholars that began in 2010 and who enrolled into a post-baccalaureate program within three years of program completion.
- Data from Integrated Postsecondary Education Data System (IPEDS) were obtained to identify staff members for all 143 institutions. Data was computed to find percentage of underrepresented racial minority faculty.
- Data from Carnegie Classification of Higher Education was used to find the research levels for R1 schools.
- Research level one (R1) Universities, which has the highest level of research activities and awarded at least 20 research/scholarship doctoral degree for this year. The research levels are based on the Carnegie classification of research levels.
- Success rates were calculated by looking at the starting cohort size in 2010 and identifying numbers of scholars enrolled in a post-baccalaureate program three years after completing the program. Then it was analyzed with the percentage of underrepresented faculty members at each school to find any effects that the faculty would have with the success rate of the McNair Program.



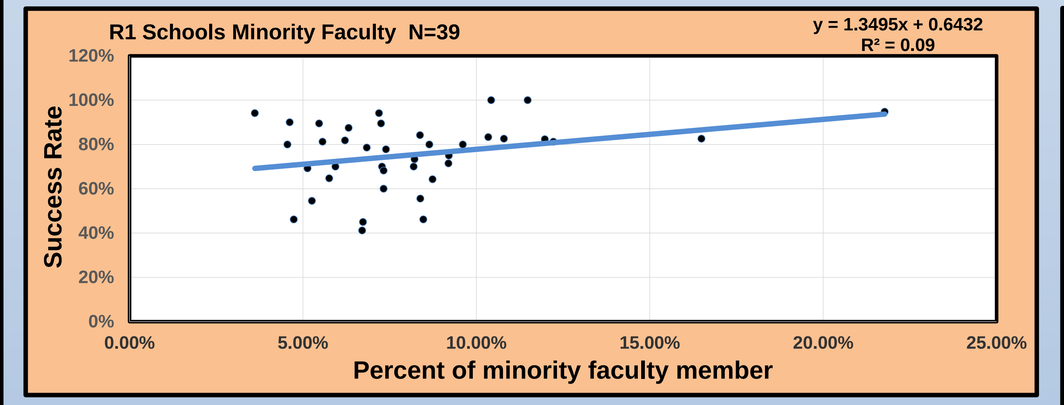
"Before you can make a dream come true, you must first have one" Ronal E. McNair

Research Question

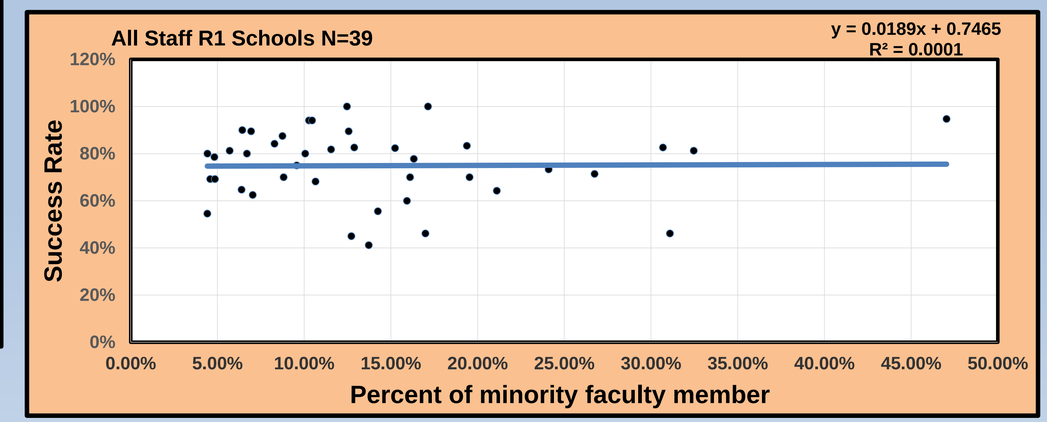
What effect does the underrepresentation of racial minority faculty at McNair host institutions have on the success rate of McNair scholars' matriculation into post-baccalaureate programs?



Graph 1:X-Y graph showing association with all underrepresented minority staff and the relation with McNair Scholars success rate



Graph 2: X-Y graph showing association with R1institutions tenure and non-tenure faculty and the relation with McNair Scholars success rate



Graph 3: X-Y graph showing association with R1 institutions all staff and the relation with McNair Scholars success rate

Results

- **Graph 1**: Among all the 143 McNair Programs locations there is no statistically significant effect of the percentage of underrepresented faculty members on McNair Scholar Success Rates.
- **Graph 2**: For the subset of McNair Program locations at 39 R1 institutions, there is a strong positive relationship between increases in the percentage of minority faculty and McNair Scholar Success rates. (R-Square= 0.09; F= 3.65, P=.06). A one-percentage point increase in the percentage of underrepresented faculty members is associated with a 1.50 percentage point increase in the success rate of the McNair Scholars.
- **Graph 3**:For the subset of McNair program locations at 39 Research 1 institutions, there was no statistically significant effect on the success on McNair Scholars when there were more minority staff at the institutions.

Discussion

This exploratory research focused on the correlation of minority staff at McNair host institutions and the effect it would have on McNair scholar success rates. Limited research has been conducted on McNair programs beyond the federally mandated annual performance reports required of each program. Data used does not take into account or reflect the following considerations:

- cultural congruency of faculty mentor pairing unknown
- program structure (academic year/summer/summer & academic year)
- socioeconomic considerations of mentors and scholars (first generation/low income)

Future Direction

Ample opportunities exist for further research in this area, particularly as it relates to underrepresented minorities persistence in STEM fields and a more systematic approach that incorporates socio economic data. Additionally, future research on impact of institution type beyond R1s is needed.

Selected References

"Find Your College." The Integrated Postsecondary Education Data System. Accessed August 03, 2016. http://nces.ed.gov/ipeds/.
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